Predictive Modeling Final Report

DSE6111

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Executive Summary

Introduction

ABC Corporation wants to determine which customers are at a high risk of attrition as well as identify key customer features that most influence this risk determination. This determination will be made using predictive modeling techniques and basic domain knowledge along with customer data provided by the company.

Objective

The primary objective of this project is to develop a model that predicts the risk of customer attrition. Using this model, ABC Corporation will be able to identify high-risk customers and mitigate the risk of attrition using targeted marketing and customer service strategies. A secondary outcome will be the identification of key customer attributes that highly influence the attrition risk.

Methodology

Several predictive models were developed, analyzed, and compared for accuracy, negative and positive result rates, interpretability, and potential economic impact. An optimal model was selected, and business recommendations were developed.

Key Findings

After developing, analyzing, and comparing six (6) predictive models, the *xgBoost model* has been selected due to its high accuracy, its solid capability to identify at-risk customers, and its low chance of misidentifying non-risk customers as customers at risk of leaving the company.

Additionally, consistent with other models, the xgBoost model solidified the identification of *Total_Trans_Ct* and *Total_Trans_Amt* features as the most influential variables from the provided customer data set.

Result

As a result of selecting the xgBoost model, ABC Corporation will be able to confidently identify at-risk customers and design programs to avert the loss of business.

Conclusion

Using the customer data provided, an xgBoost boosted decision tree model was developed and was selected as the preferred predictive model to use to meet the goals of the Company.

Approaches and Data

Overall Approach

The six models were developed using a pattern, specifically *tidymodels* recipes and workflows, that provided a framework to make the models and the output consistent and comparable to one another. Doing so facilitated the process of choosing the optimal model for ABC Corporation.

Customer Dataset

ABC Corporation provided a customer dataset that included 21 datapoints for 10,127 customer records. The dataset was reviewed in Excel for completeness, breadth and depth of data. There were no records that were missing some or all feature values, and all the data in each field appeared appropriately consistent in type and content. The dataset was scanned for records to be removed prior to the modeling exercises, but none were found. However, there was one feature that would need to be eliminated as described in the next section.

Feature Engineering and Model-agnostic Processing Steps

Variable removed from data set

The CLIENTNUM variable serves as an identifier for the customer. This information would not play a role in the modeling exercises, so this variable was removed.

Inclusion of all variables

It is important to note that all variables, quantitative and qualitative, were used in the development of the models.

Identification of training and test data subsets

The customer data was split into a training subset that contained 75% of the full data set and a test subset that contained the remaining 25%.

Cross-validation

All models included cross-validation of 10 folds to improve accuracy and reduce the risk of overfitting.

Metrics captured

The following metrics were captured as output from each model:

- Accuracy
- Counts of true negative, true positive, false negative and false positive predictions
- Rates of true positives and false positives
- Area under the curve as it relates to the ROC curve

Influential features

The models provided data about the influence of each variable. As noted in the comparison section, the models consistently identified the most important features.

Plots

The following plots were captured for each model:

- Feature importance bar chart
- ROC using the last fit model
- ROC calculation breakdown using the selected threshold

Thresholds

Each ROC calculation was evaluated using thresholds of .25, .45, .6, .75, and .9.

null model

The null model was specifically compared against each of the other models as part of the evaluation.

Predictive Modeling Methods Used

For this project, six predictive models were developed and analyzed. Each of these models has properties that are suitable for the scenario being addressed, as well as properties that are not appropriate nor supportive.

The first model is the null model. This model assumes that there is no relationship between variables and is a useful baseline to compare all models against. If other models perform better than the null model, this outcome suggests that those models have successfully learned using some or all the variables in the training data. (Eskandar, 2023)

The second model is the logistic regression model, specifically using the lasso technique. The logistic regression model is commonly used to provide predictions with values between 0 and 1 for classification purposes. It is one of the more easily applied models and is relatively easy to interpret. It does have the disadvantage of keeping all variables in play, even the least important ones. The lasso technique reduces the risk of incorporating non-impactful variables by adding a penalty to those variables, which in turn makes those variables zero. (Awati, 2017)

The third model is the kNN (K-nearest neighbors) model assigns a new observation to a class based on proximity between new observation and other data. This model can be used for any shape (linear, non-linear) of data. The advantage of this model is that it is easy to understand but requires large datasets and requires identifying an acceptable distance between observation and class. (Varghese, 2018) This model does not directly identify feature importance.

The fourth model is the naïve Bayes model, based on Bayes' Theorem. This classification model assumes that each feature is independent of any another. (Ray, 2024) This model does not directly identify feature importance, although this data was developed using an alternative in this project.

The fifth model is the random forest model. This model combines the output of several decision trees created in parallel to arrive at a single prediction. (Donges & Whitfield, 2024) This model builds trees based on decisions between features using Gini impurity. This model is often used to identify feature importance.

The final model is the xgBoost model. This model is a specific type of boosted decision tree model that creates sequential decision points and builds off the prior decisions to improve the model. (Ashtari, 2024) Gini impurity is used in the decision process. This model can identify feature importance.

Detailed Findings and Evaluation

Compare thresholds and metrics within each model

null Model

The null model developed for this data set resulted in an accuracy metric of 84% and an area under the ROC curve of 50%

```
> null_metrics
# A tibble: 3 × 6
  .metric     .estimator  mean     n std_err .config
  <chr>       <chr>      <dbl> <int>   <dbl> <chr>
1 accuracy    binary     0.839    10 0.00369 Preprocessor1_Model1
2 brier_class binary     0.135    10 0.00250 Preprocessor1_Model1
3 roc_auc     binary     0.5      10 0       Preprocessor1_Model1
```

Figure 1: null Model metrics

Logistic regression with lasso

This model produced an accuracy rate of 84% and an AUC of 80% against the ROC. The resulting ROC curve was only moderately impressive.

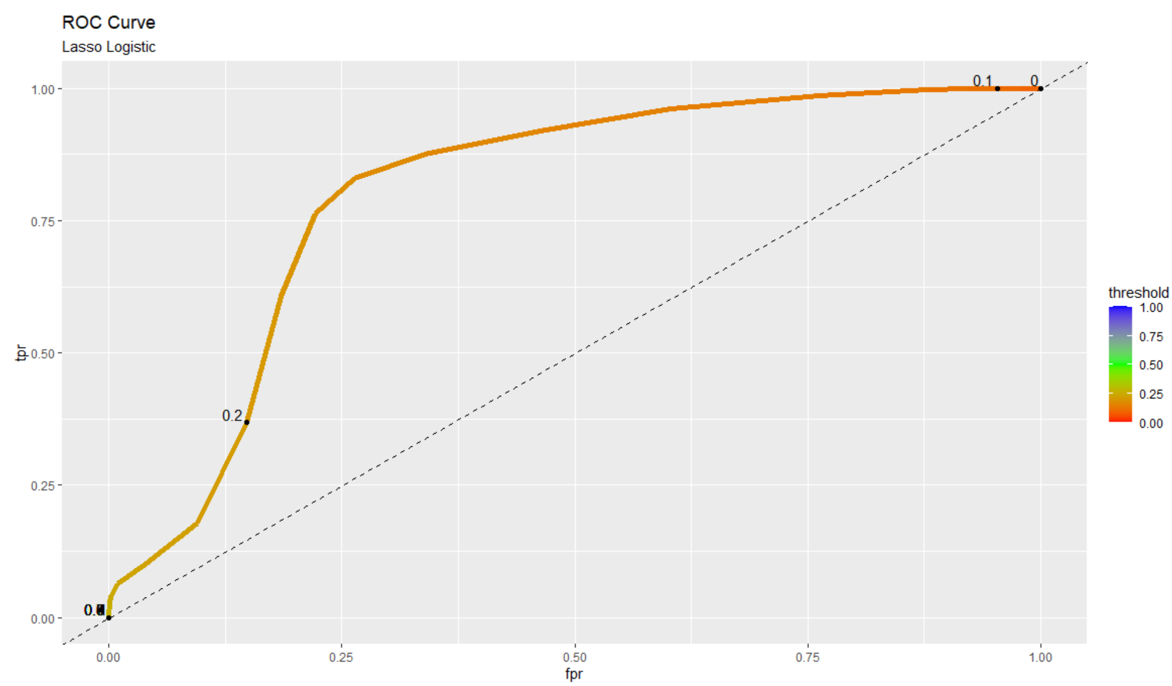

Figure 2: ROC curve from logistic regression with lasso model

Comparing the effect of a variety of thresholds on the ROC curve calculation produced the following outputs

	Full Logistic	Full Logistic	Full Logistic	Full Logistic	Full Logistic
Threshold Used	**0.23**	0.45	0.6	0.75	0.9
tn	**2104**	2125	2125	2125	2125
fn	**381**	407	407	407	407
fp	**21**	0	0	0	0
tp	**26**	0	0	0	0
total	**2532**	2532	2532	2532	2532
fpr = fp/(fp + tn)	**0.0099**	0.0000	0.0000	0.0000	0.0000
tpr = tp/(tp + fn)	**0.0639**	0.0000	0.0000	0.0000	0.0000
model accuracy	**0.8412**	0.8393	0.8393	0.8393	0.8393
model roc_auc	**0.803**	0.803	0.803	0.803	0.803
Performance against null: accuracy (mean)	**0.839**	0.839	0.839	0.839	0.839
Performance against null: roc_auc (mean)	**0.5**	0.5	0.5	0.5	0.5

Figure 3: Effect of threshold changes on TPR, FPR, and ROC curve calculation

The model required a very low threshold to produce reasonably accurate levels of positive and negative predictions. While the FPR was very low, well below 1%, the TRP was approximately 6%, which is also very low. The null model had a similarly accurate result.

Another view of this threshold effect can be seen in this plot, where there is a significant drop-off in the number of positive predictions after regularization reaches .15 or so.

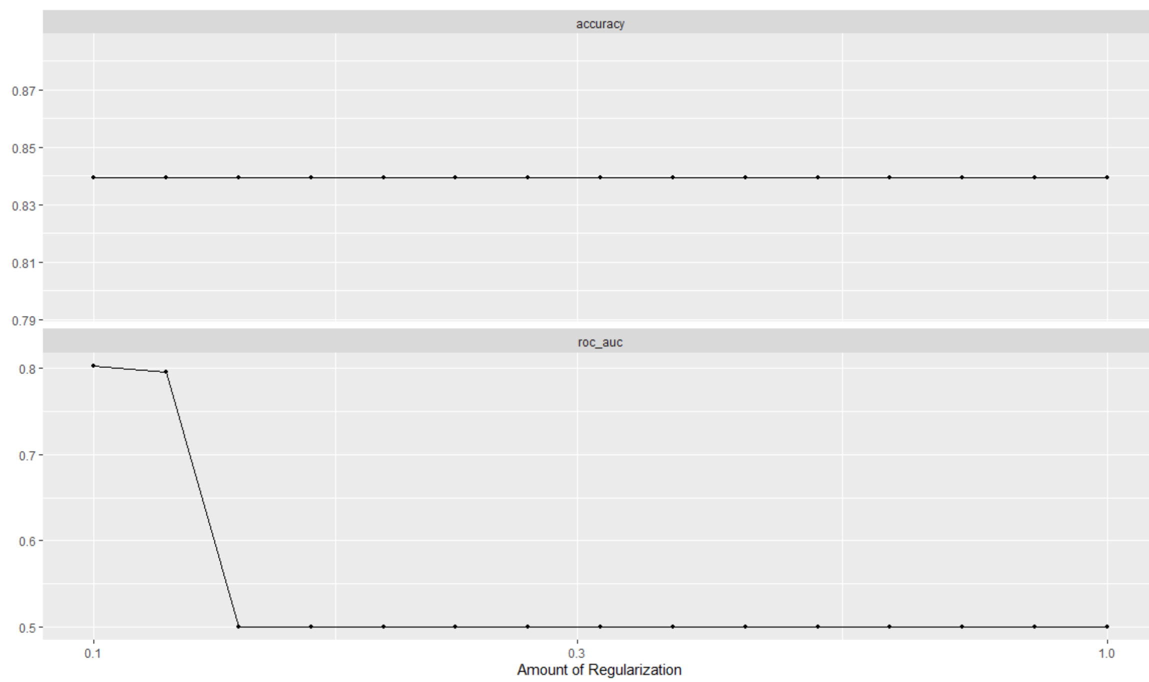

Figure 4: Effect of increased regularization against accuracy and ROC_AUC

After choosing .23 as the most useful threshold for purposes of this project, the ROC curve calculation and plotting still had only a few positive data points.

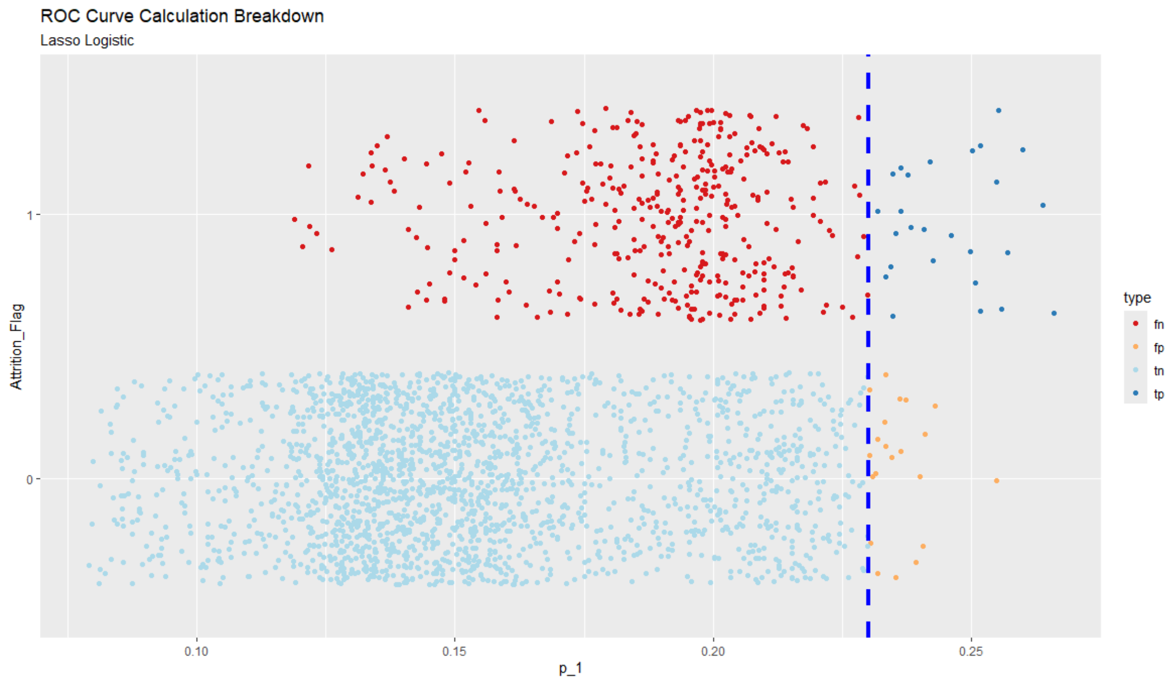

Figure 5: ROC Curve calculation for logistic regression at selected threshold

The logistic model with lasso identified this feature importance bar chart, sorted by level of importance. The Total_Trans_Ct and Total_Trans_Amt variables are the most influential variables of this model.

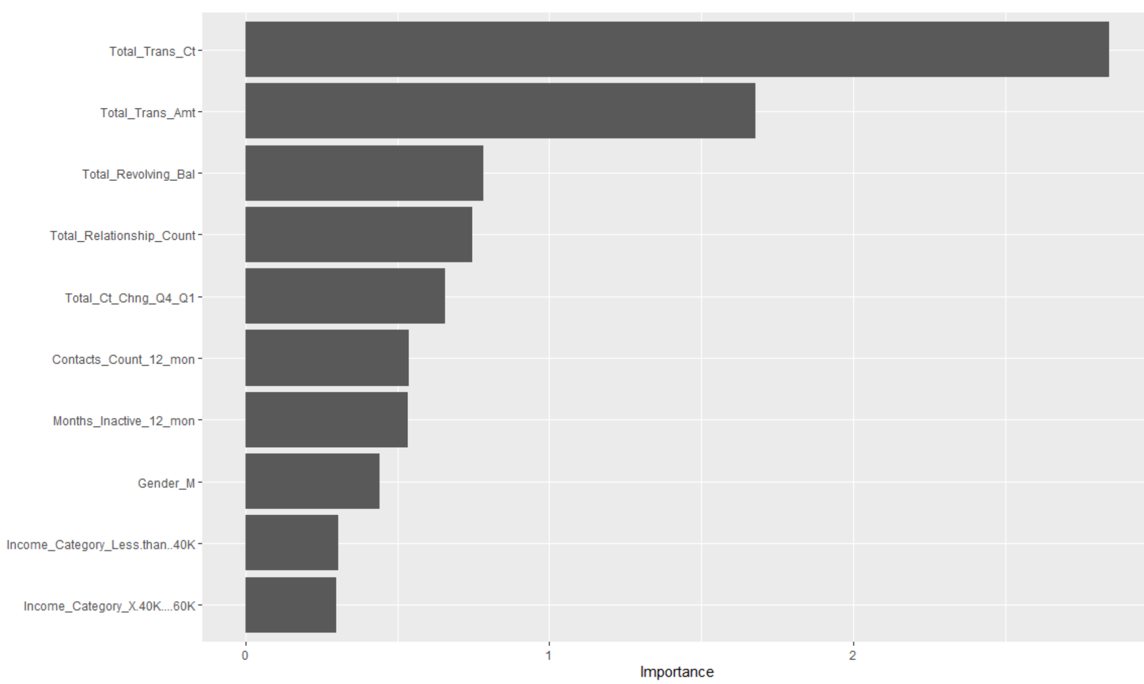

Figure 6: Logistic regression feature importance chart

kNN

This model produced an accuracy rate of 88% and an AUC of 86% against the ROC. The resulting ROC curve was reasonably shaped but still did not have a large area under the curve.

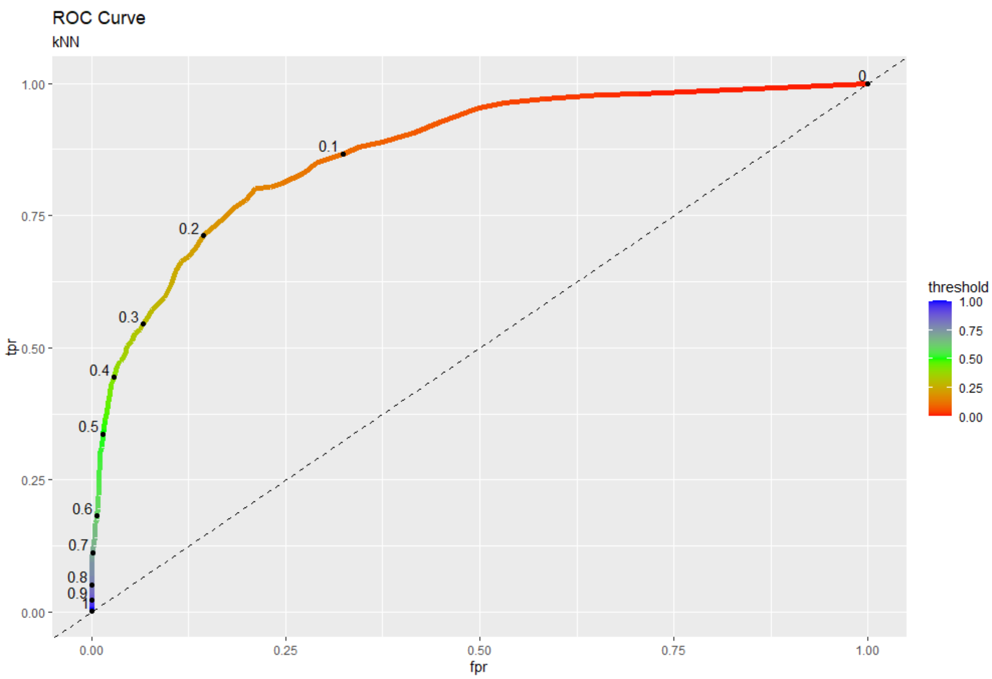

Figure 7: ROC curve from kNN model

Comparing the effect of a variety of thresholds on the ROC curve calculation produced the following outputs

	Full kNN	Full kNN	Full kNN	Full kNN	Full kNN
Threshold Used	0.25	**0.45**	0.6	0.75	0.9
tn	1908	**2079**	2109	2113	2125
fn	155	**247**	333	244	398
fp	217	**46**	16	12	0
tp	252	**160**	74	163	9
total	2532	**2532**	2532	2532	2532
fpr = fp/(fp + tn)	0.1021	**0.0216**	0.0075	0.0056	0.0000
tpr = tp/(tp + fn)	0.6192	**0.3931**	0.1818	0.4005	0.0221
model accuracy	0.8531	**0.8843**	0.8622	0.8989	0.8428
model roc_auc	0.864	**0.864**	0.864	0.953	0.953
Performance against null: accuracy (mean)	0.839	**0.839**	0.839	0.839	0.839
Performance against null: roc_auc (mean)	0.5	**0.5**	0.5	0.5	0.5

Figure 8: Effect of threshold changes on TPR, FPR, and ROC curve calculation

The accuracy and ROC_AUC did not vary much as the thresholds were tested. Like the logistic regression model, the FPR was very low at 2% and the TPR was also low at 39%. The overall level of accuracy and other metrics did not vary much from the null model. This can also be seen in the plot below.

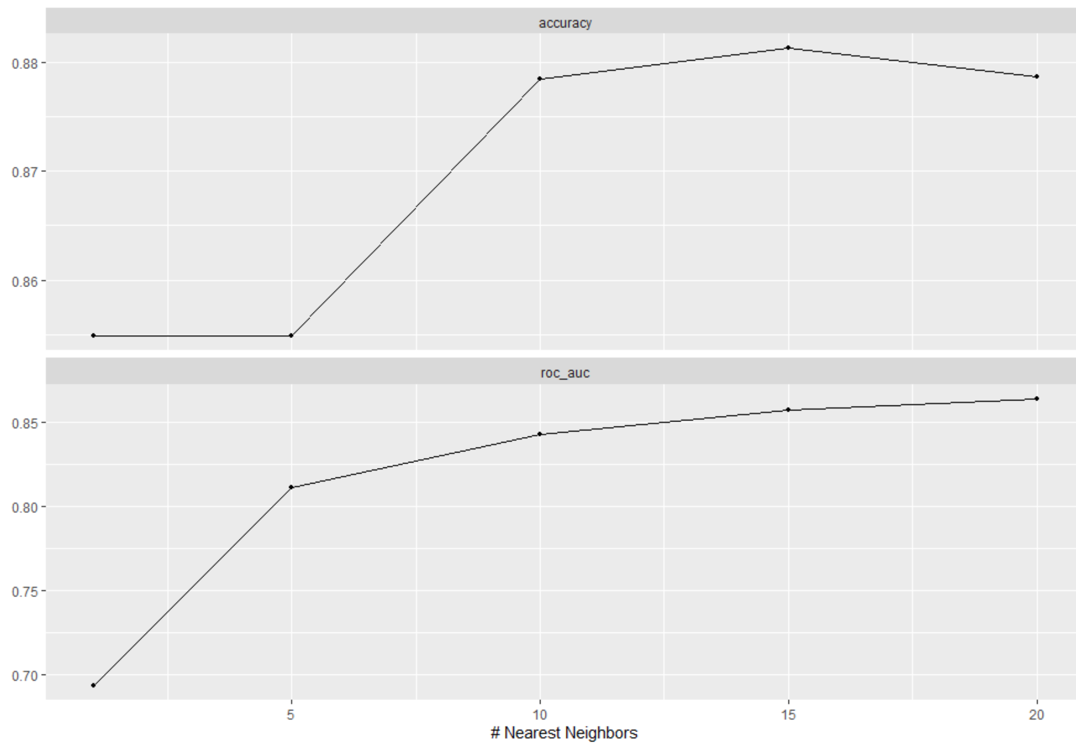

Figure 9: Effect of increased regularization against accuracy and ROC_AUC

Choosing .45 as the most useful threshold for purposes of this project, the ROC curve calculation and plotting had a variety of true and false positive and negative predictions.

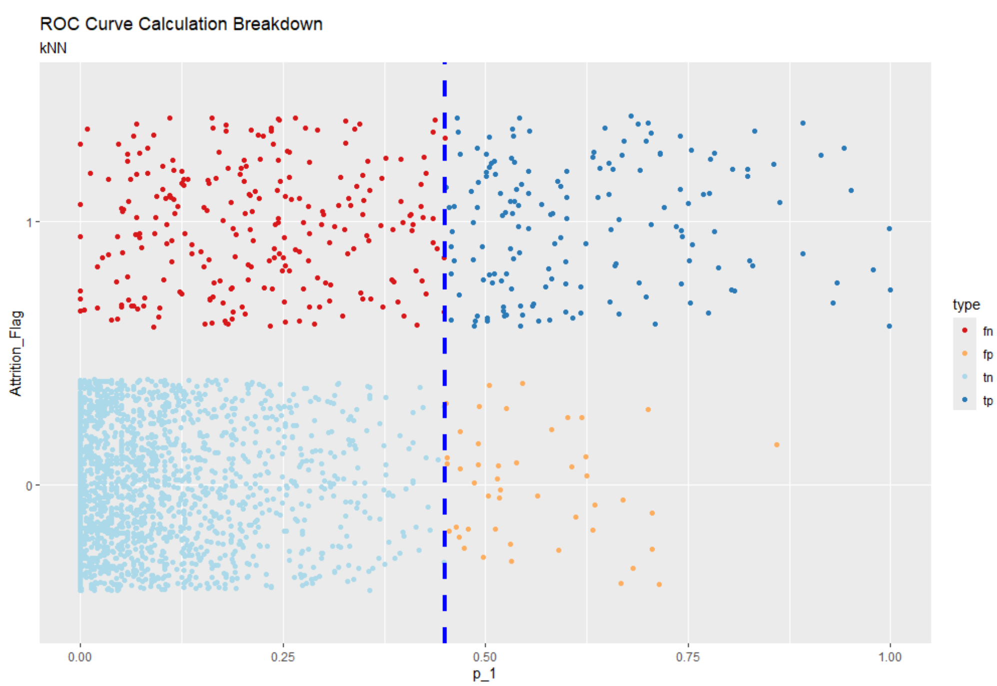

Figure 10: ROC Curve calculation for kNN model at selected threshold

No feature importance output was generated from this model, primarily due to the lack of inherit capability within the model itself.

Naïve Bayes model

This model produced an accuracy rate of 85% and an AUC of 95% against a strangely shaped ROC curve. Overall, there was still not a large area under the curve.

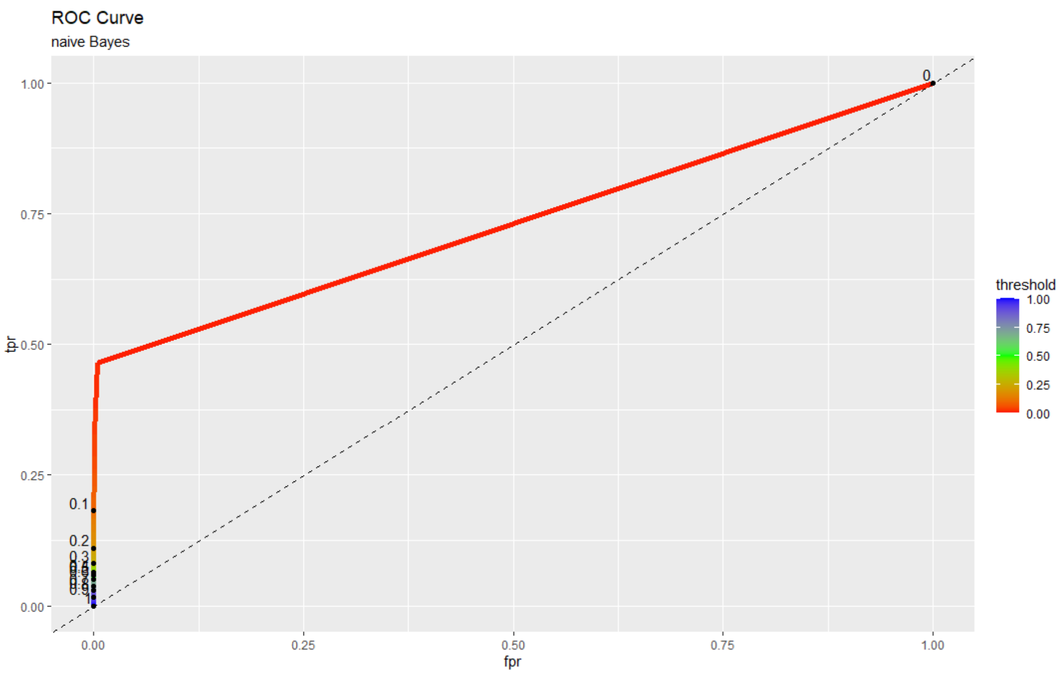

Figure 11: ROC curve from naive Bayes model

Comparing the effect of a variety of thresholds on the ROC curve calculation produced the following outputs

	full naïve Bayes	full naïve Bayes	full naïve Bayes	full naïve Bayes	full naïve Bayes
Threshold Used	**0.25**	0.45	0.6	0.75	0.9
tn	**2125**	2125	2125	2125	2125
fn	**372**	382	386	392	400
fp	**0**	0	0	0	0
tp	**35**	25	21	15	7
total	**2532**	2532	2532	2532	2532
fpr = fp/(fp + tn)	**0.0000**	0.0000	0.0000	0.0000	0.0000
tpr = tp/(tp + fn)	**0.0860**	0.0614	0.0516	0.0369	0.0172
model accuracy	**0.8531**	0.8491	0.8476	0.8452	0.8420
model roc_auc	**0.964**	0.964	0.964	0.964	0.964
Performance against null: accuracy (mean)	**0.839**	0.839	0.839	0.839	0.839
Performance against null: roc_auc (mean)	**0.5**	0.5	0.5	0.5	0.5

Figure 12: Effect of threshold changes on TPR, FPR, and ROC curve calculation

The model required a very low threshold to produce reasonably accurate levels of positive and negative predictions. There were no false positives at this threshold, but the TPR was low at 8%. The null model had a similarly accurate result.

After choosing .25 as the most useful threshold for purposes of this project, the ROC curve calculation and plotting still had only a few positive data points. This was like, albeit slightly better than, the logistic model.

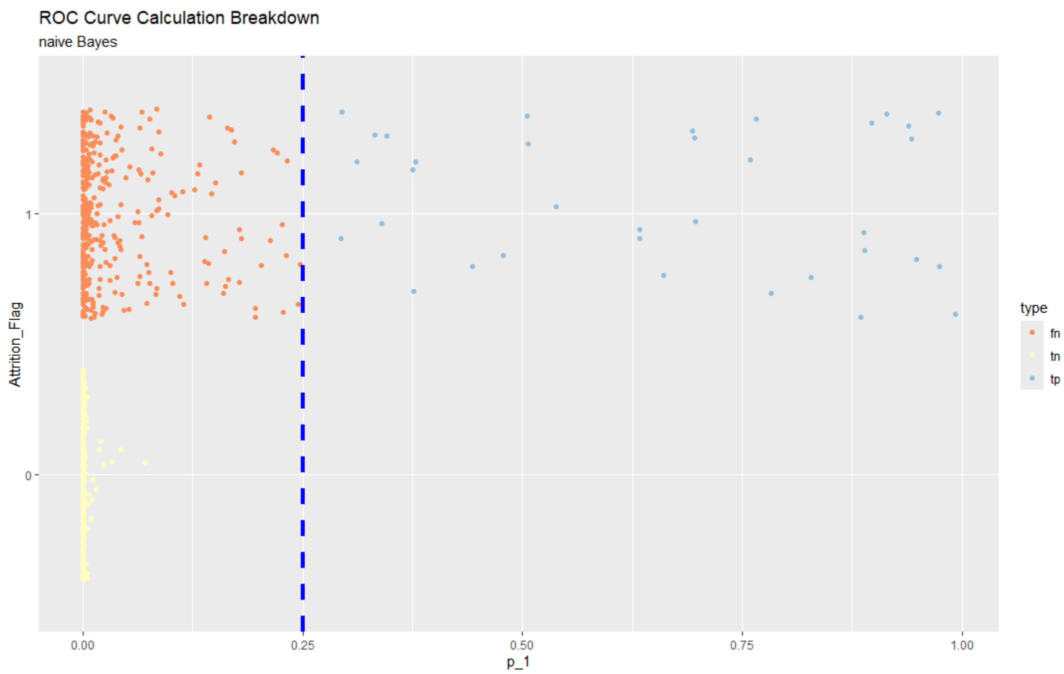

Figure 13: ROC Curve calculation for naive Bayes model at selected threshold

The naïve Bayes model identified this feature importance table, sorted by level of importance. The Total_Trans_Ct and Total_Ct_Chng_Q4_Q1 variables are the most influential variables of this model. Interestingly, the Total_Trans_Amt was less important in this model.

```
> varImp(impModel)
ROC curve variable importance

                          Importance
Total_Trans_Ct                100.00
Total_Ct_Chng_Q4_Q1            84.35
Avg_Utilization_Ratio          66.20
Total_Trans_Amt                62.35
Contacts_Count_12_mon          53.52
Months_Inactive_12_mon         41.45
Total_Relationship_Count       31.88
Total_Amt_Chng_Q4_Q1           27.19
Months_on_book                  0.00
```

Figure 14: Feature Importance Table for naive Bayes model

Random Forest

This model produced a ROC curve that had substantial area under the ROC curve. The model's accuracy rate was a much-improved 94% and the AUC under the ROC curve improved to 98%.

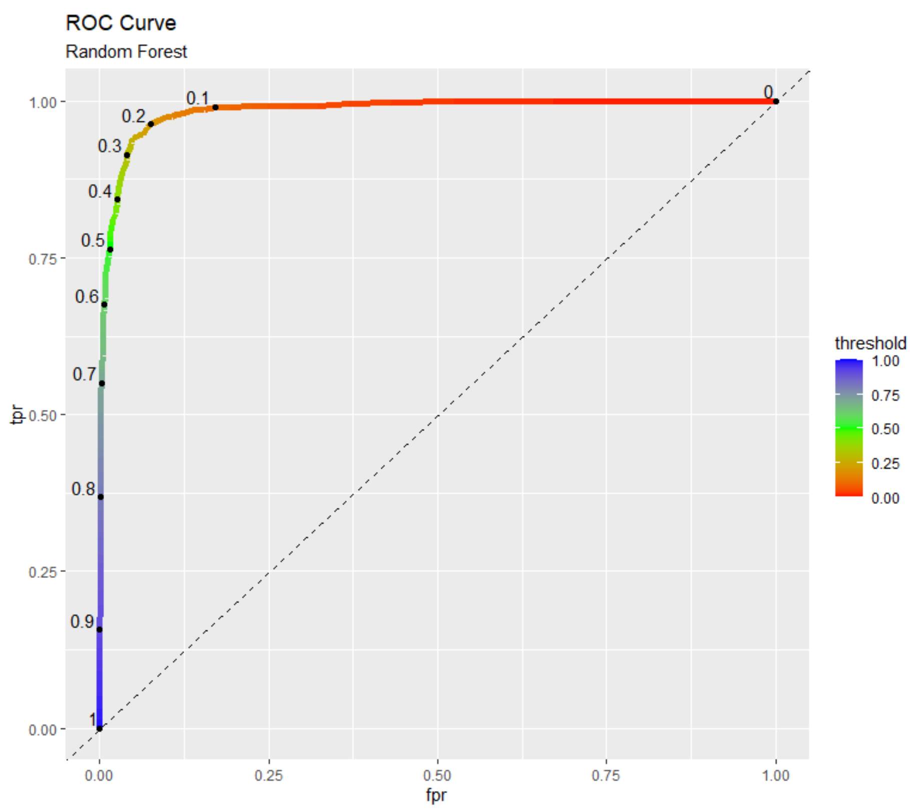

Figure 15: ROC curve from Random Forest model

Comparing the effect of a variety of thresholds on the ROC curve calculation produced the following outputs

	Full random forest	Full random forest	Full random forest	Full random forest	random forest
Threshold Used	0.2	0.45	**0.6**	0.75	0.9
tn	2000	2083	**2110**	2123	2122
fn	22	78	**132**	220	258
fp	125	42	**15**	2	3
tp	385	329	**275**	187	149
total	2532	2532	**2532**	2532	2532
fpr = fp/(fp + tn)	0.0588	0.0198	**0.0071**	0.0009	0.0014
tpr = tp/(tp + fn)	0.9459	0.8084	**0.6757**	0.4595	0.3661
model accuracy	0.9419	0.9526	**0.9419**	0.9123	0.8969
model roc_auc	0.988	0.988	**0.988**	0.988	0.988
Performance against null: accuracy (mean)	0.839	0.839	**0.839**	0.839	0.839
Performance against null: roc_auc (mean)	0.5	0.5	**0.5**	0.5	0.5

Figure 16: Effect of threshold changes on TPR, FPR, and ROC curve calculation

The model required a moderate threshold of .6 to produce reasonably accurate levels of positive and negative predictions. In this case the false positive rate was less than 1% and the true positive rate was a moderate 68%. This model performed better than the null model.

The accuracy and ROC_AUC did not vary much as the thresholds were tested.

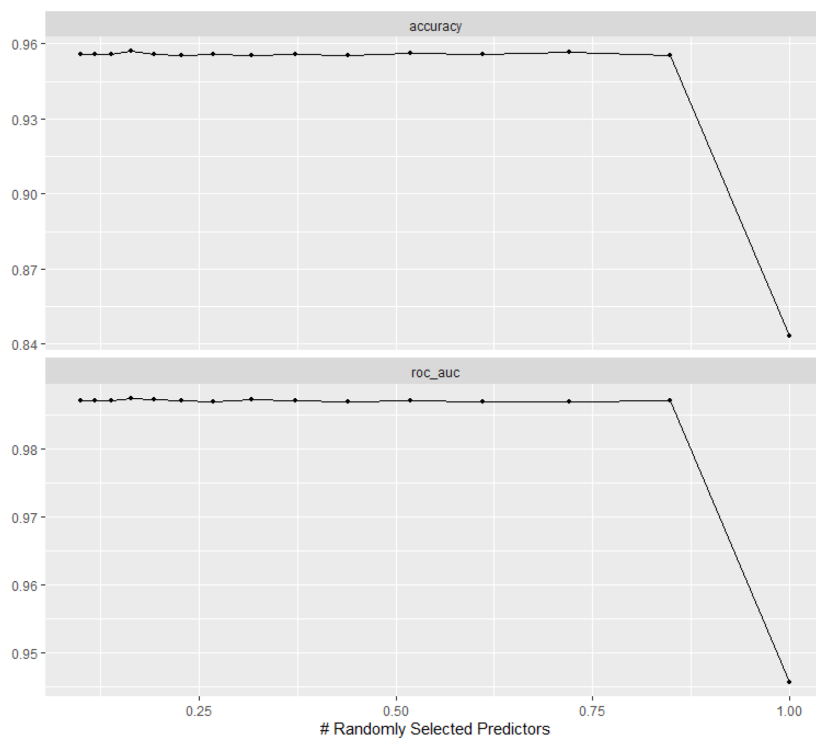

Figure 17: Effect of increased regularization against accuracy and ROC_AUC

After choosing .6 as the most useful threshold for purposes of this project, the ROC curve calculation and plotting had a variety of true and false positive and negative predictions.

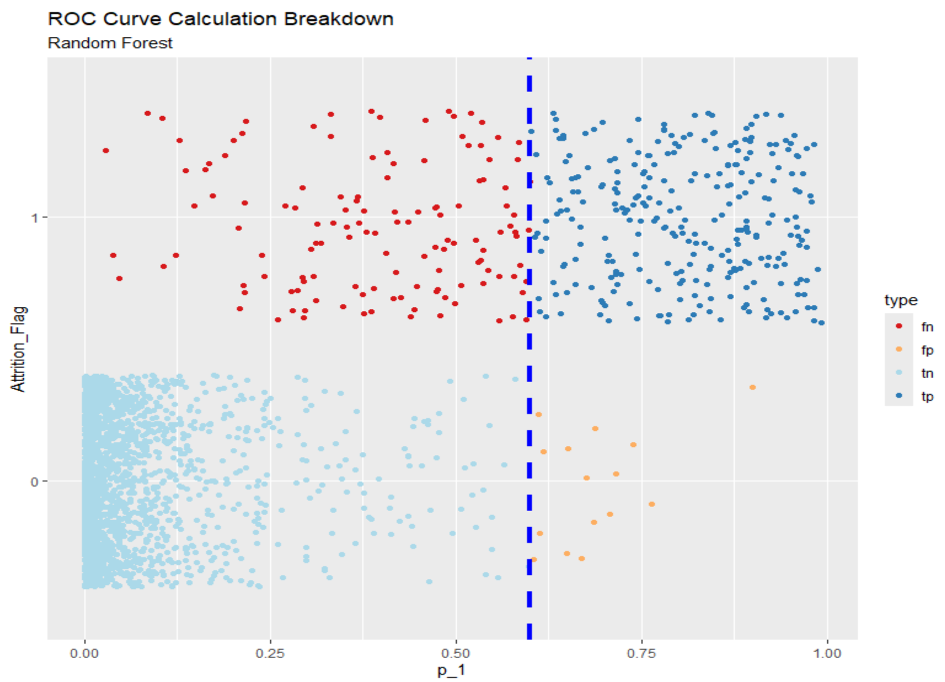

Figure 18:L ROC Curve calculation for the Random Forest model at selected threshold

The random forest model identified this feature importance bar chart, sorted by level of importance. The Total_Trans_Amt and Total_Trans_Ct variables are the most influential variables of this model.

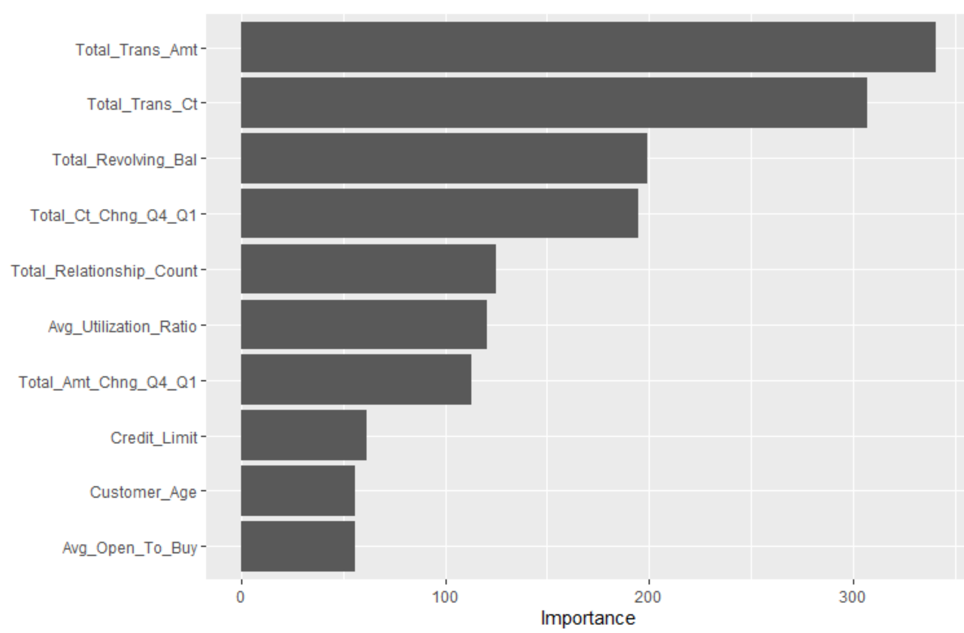

Figure 19: Random forest feature importance chart

xgBoost

This model produced a ROC curve with a very significant amount of area under the ROC curve. The model is highly accurate with a 96% accuracy rate. The area under the ROC curve is excellent at 99%.

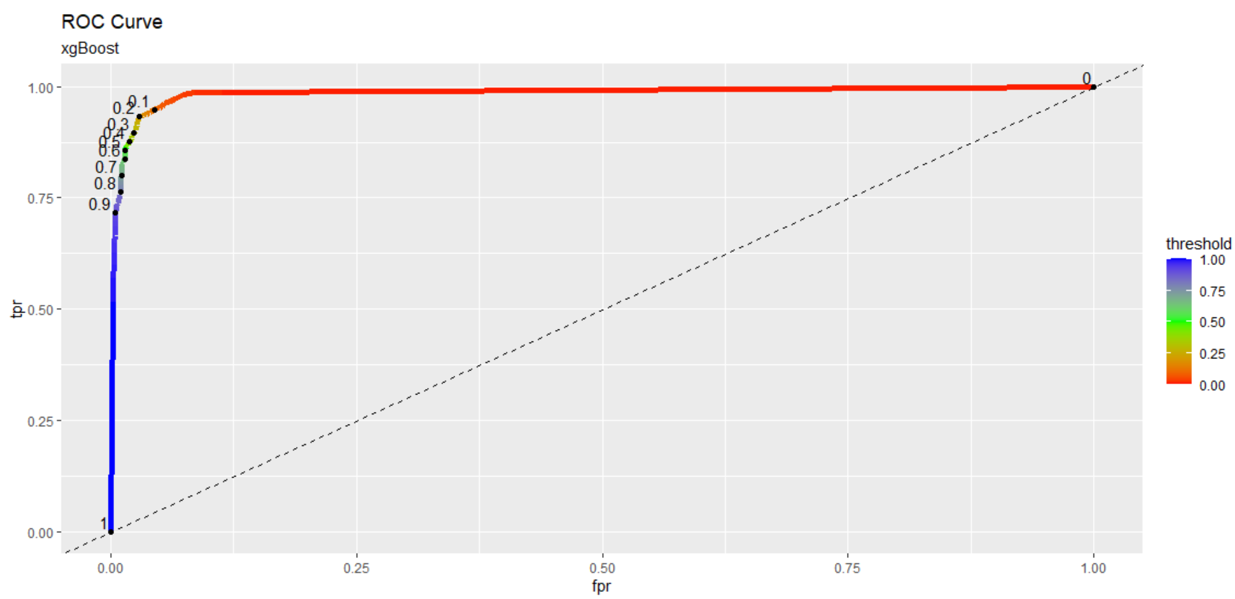

Figure 20: ROC curve from the xgBoost model

Comparing the effect of a variety of thresholds on the ROC curve calculation produced the following outputs

	xgBoost	xgBoost	xgBoost	xgBoost	xgBoost
Threshold Used	0.25	0.45	0.6	0.75	**0.9**
tn	2078	2093	2103	2107	**2116**
fn	37	46	59	75	**91**
fp	47	32	22	18	**9**
tp	370	361	348	332	**316**
total	2532	2532	2532	2532	**2532**
fpr = fp/(fp + tn)	0.0221	0.0151	0.0104	0.0085	**0.0042**
tpr = tp/(tp + fn)	0.9091	0.8870	0.8550	0.8157	**0.7764**
model accuracy	0.9668	0.9692	0.9680	0.9633	**0.9605**
model roc_auc	0.991	0.991	0.991	0.991	**0.991**
Performance against null: accuracy (mean)	0.839	0.839	0.839	0.839	**0.839**
Performance against null: roc_auc (mean)	0.5	0.5	0.5	0.5	**0.5**

Figure 21: Effect of threshold changes on TPR, FPR, and ROC curve calculation

A very high threshold of .9 produced very accurate levels of positive and negative predictions with a false positive rate of less than 1% and a good TPR of 77.6%. This model performed significantly better than the null model.

Another view of this threshold effect can be seen in these plots, there is little variation in the area under the ROC curve as the learn_rate, tree_depth, and number of trees change.

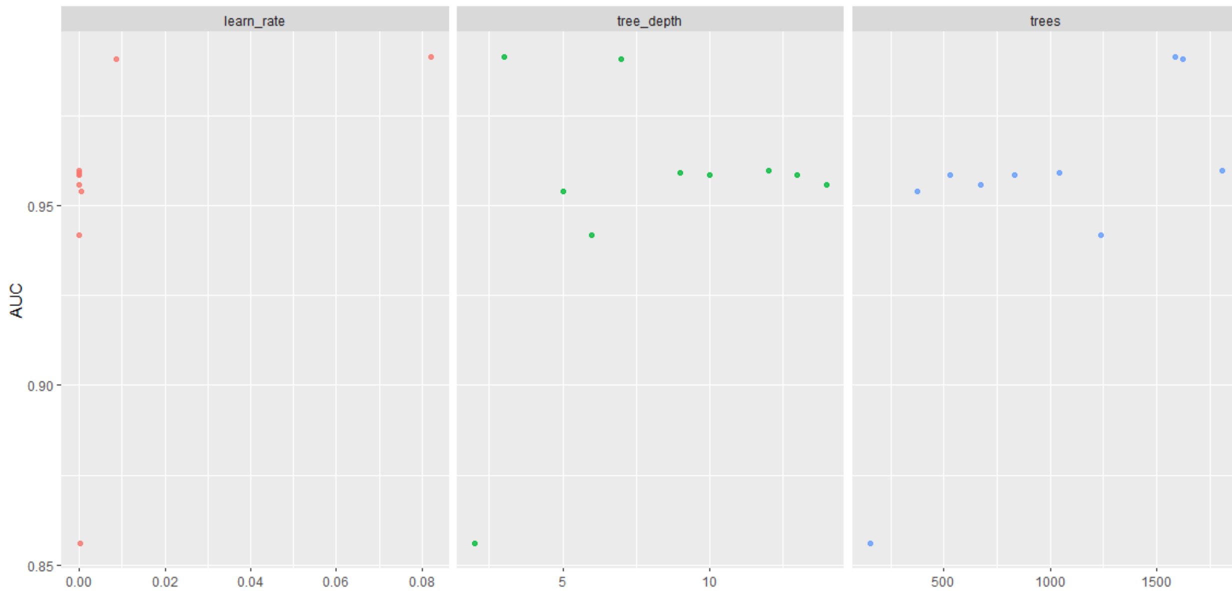

Figure 22: Effect of learn_rate, tree_depth, and number of trees on the area under ROC curve.

After choosing .9 as the most useful threshold for purposes of this project, the ROC curve calculation and points had a very reasonable amount of true and false positive and negative predictions.

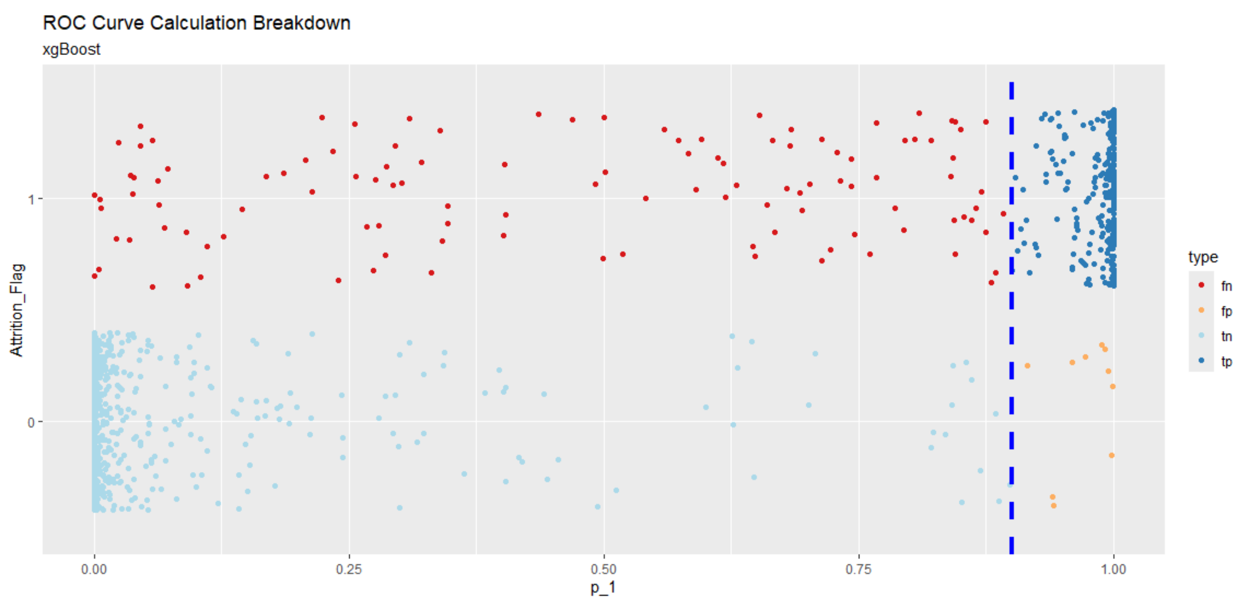

Figure 23: ROC Curve calculation for xgBoost model at selected threshold

The xgBoost model identified this feature importance bar chart, sorted by level of importance. The Total_Trans_Ct and Total_Trans_Amt variables are the most influential variables of this model.

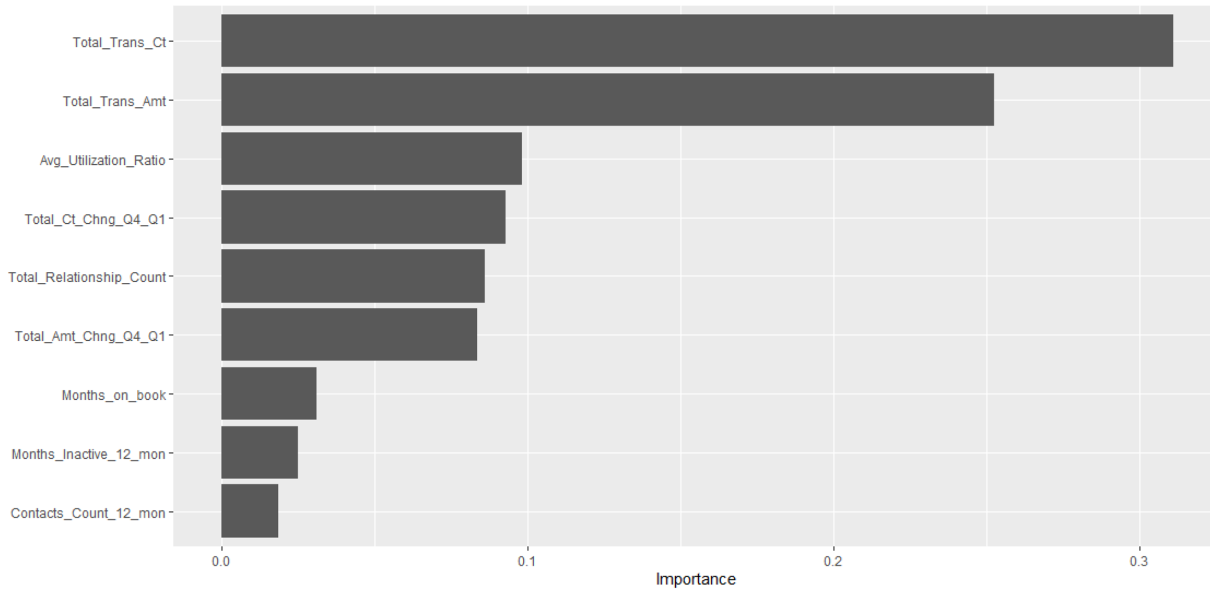

Figure 24 xgBoost feature importance chart

Compare metrics between selected threshold/model

This table was used to compare key metrics of these models when the "best" threshold values are applied to the models.

	Full Logistic	Full kNN	Full naïve Bayes	Full random forest	Full xgBoost
Threshold Used	0.23	0.45	0.25	0.6	0.9
tn	2104	2079	2125	2110	2116
fn	381	247	372	132	91
fp	21	46	0	15	9
tp	26	160	35	275	316
total	2532	2532	2532	2532	2532
fpr = fp/(fp + tn)	0.0099	0.0216	0.0000	0.0071	0.0042
tpr = tp/(tp + fn)	0.0639	0.3931	0.0860	0.6757	0.7764
model accuracy	0.8412	0.8843	0.8531	0.9419	0.9605
model roc_auc	0.803	0.864	0.952	0.988	0.991
Performance against null: accuracy (mean)	0.839	0.839	0.839	0.839	0.839
Performance against null: roc_auc (mean)	0.5	0.5	0.5	0.5	0.5

Figure 25: Comparison of key metrics between models at specific threshold values

Referencing this table makes selecting the best performing model in the context of this company's business needs easier.

Models Not Selected

The logistic regression model with lasso, kNN model, naïve Bayes model, and the random forest model were not selected as the optimal model for this company's business needs. Each of these models had pitfalls. Logistic regression with lasso and the naïve Bayes models performed on par with the null model, so it was hard to think of these two models as value-added options. The kNN model didn't automatically provide a way to identify key features, which would have resulted in either additional work to pull something together or fail to address one of the Company's specific requests. The random forest model became a plausible second choice option and should be kept in mind if similar requests come in future.

Model Selected

The xgBoost model was selected as the optimal model for this company's requirements. The model performed about as well as one could hope, and didn't take very long to process the customer data to obtain a very accurate prediction. This model performed much better than the null model. Also, the model produced only 100 false positives and false negatives, which will hopefully make the job of the customer service department's jobs easier by not mis-targeting too many people. This model also had the highest number of true positive predictions. The xgBoost model also produced a feature importance ranking, with similar results as the logistic and naïve Bayes models.

Feature Importance Discussion

ABC Corporation has also requested identification of the customer characteristics that appear to most influence the risk of attrition. If ABC Corporation knows what the characteristics are, it can design better targeted marketing efforts that impact those characteristics, hopefully reducing the attrition risk.

Total_Trans_Ct and Total_Trans_Amt

Models that produce a feature importance list are identified below and given a number ranking based on the order of ranking identified in each feature importance list. The top 2 features are also highlighted for visibility ease.

	Order per each model:					
Feature Importance List	**Logistic**	**kNN**	**naïve Bayes**	**random forest**	**xgBoost**	**Count**
Total_Trans_Ct	1		1	2	1	4
Total_Trans_Amt	2		4	1	2	3
Total_Ct_Chng_Q4_Q1	4		2	3	4	1
Avg_Utilitzation_Ratio	6		3	4	3	
Total_Relationship_Count	3		7	5	5	
Months_Inactive_12_mon	5		6	8	8	
Contacts_Count_12_mon	7		5	9	9	
Months_on_book	8		9	7	7	
Total_Amt_Chng_Q4_Q1	9		8	6	6	

Figure 26: Comparative Feature Importance List

In the far-right column, the count of models with either a 1 or 2 in the feature ranking are summed. Total_Trans_Ct and Total_Trans_Amt most frequently appear as the most important features that influence the risk of attrition. Our optimal model, the xgBoost model, also identified these two variables as the most influential within that model.

Looking to support the assertion that these two variables, Total_Trans_Ct and Total_Trans_Amt, influence the risk of attrition, the predictions from the last_fit xgBoost model were bound with the test set data. The resulting data was visually reviewed in Excel at a very high level. The negative predictions (both true and false) had a range of Total_Trans_Ct counts from 13 to 131 and a range of Total_Trans_Amt values from 857 to 18484, while the positive predictions had a range of Total_Trans_Ct counts from 10 to 94 and a range of Total_Trans_Amt values from 644 to 10291. From this information, it was surmised that the higher the Total_Trans_Ct counts and Total_Trans_Amt, the lower the predicted attrition risk. This assertion will come into play as recommendations to the Company.

Contacts_Count_12_mon

This customer characteristic touches on some of the limits of the predictive modeling & business strategy recommendation process when full domain knowledge is missing.

This variable was not identified as an important feature by any of the models, but there could be some value in exploring this characteristic more deeply. While the shared definition of this characteristic is "No. of Contacts in the last 12 months", the full context is missing. Questions to be asked could include

- What is the content and messaging within the contact?
- Who initiates the contact?
- Are these communications related to
    - customer service concerns?
    - repeated requests for assistance?
    - payment collection attempts?
- What is the tone of the communications – positive, negative?
- How does the Corporation know whether a customer is satisfied or not?

Depending on the answers provided and other exploration (e.g., reviewing call logs, email messages, etc.), additional recommendations could be developed and applied. For instance, if the Corporation is reaching out to the customer only when the customer's payment is overdue or when the customer has lodged a customer service complaint, then there are probably opportunities for the Corporation to adjust their overall customer communication strategy to have a positive, proactive, cordial tone.

Recommendations

To support ABC Corporation's effort to reduce customer attrition, the following recommendations have been developed. Interpreting and incorporating these recommendations require care and viewing them through the lens of the precise corporate customer base, industry, and economic situation.

Recommendation 1: Increase transaction counts and amounts

The two most important features of this data set are the Total_Trans_CT and Total_Trans_Amt features, and further analysis validated a theory that the more transaction counts and increased total annual transaction amounts, the lower the risk of attrition. Therefore, the primary recommendation is to increase the count and value of transactions.

This can be accomplished in several ways, including a loyalty program that rewards customers as the number of transactions or the total annual value of transactions increase. To implement this program, the corporation would need to allocate a small budget for rewards, as well as budget for setting up, marketing, and maintaining the program. Depending on what the initial budget is, the corporation could implement such a program just for the customers with attrition risk scores greater than the threshold amount identified in the xgBoost model if they want to keep the program small at first. Alternatively, the corporation could institute the program for all customers. In either case, success with a loyalty program would probably mean an overall reduced attrition rate for the most at-risk customers.

Another method of increasing the number of transactions or the total annual value of transactions might be to extend special offers specifically to the at-risk customers. However, it would be wise to get a good understanding of these customers' level of satisfaction with the company and what could be done to improve that level.

Recommendation 2: Get to know these customers better

While the data in the dataset is relatively clear of bad/messy data, missing values, inconsistent values, it doesn't tell the entire story. Data that captures customer sentiment, values, perspectives, and reasons for choosing the Corporation would be helpful to get a more complete picture of the customer base. Gathering this data requires an intentional effort and might be something that the company outsources if in-house competency doesn't exist.

Examples of ways to get to know the customer better include surveys and customer touch-base calls or emails. Both methods require well thought out organization and communication details, as well as a commitment to follow up or through on any negative feedback. Questions the company needs to ask itself might include "How will we handle feedback that isn't glowing?" The company could view that information as an opportunity to make improvements and have the customer experience the improvement. Additionally, gathering feedback from customers is not a one-and-done effort. Repeated cycles of intentional communication strengthen customer relationships.

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